Technische Universität Berlin

Fakultät VII Wirtschaft und Management

Fachgebiet Management im Gesundheitswesen

Masterarbeit zum Thema:

Telemedicine Platforms as an Instrument to Improve Healthcare – Development of a Digital Patient and Doctor Portal in Uzbekistan

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Semester: Wintersemester 2025/26

Abgabedatum: 01.03.2026

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Abstract

Der Abstract ist eine Zusammenfassung deiner Masterarbeit. Du nennst die Fragestellung deiner Arbeit, beschreibst die Methode und stellst die wichtigsten Ergebnisse dar.

Mehr Informationen zum Abstract findest du unter: Den perfekten Abstract deiner Masterarbeit schreiben mit Beispiel

Inhaltsverzeichnis

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1. Introduction

1.1 Background and Rationale

Uzbekistan's health system, like those of many post-Soviet states, has undergone a complex and protracted reform process since independence in 1991. The legacy of the Semashko model - characterized by centralized planning, public financing and provision, and a strong hospital sector - has shaped both the strengths and persistent challenges of the system. Over the past three decades, the country has sought to replace input-driven norms with outcomes-focused governance, strengthen primary health care (PHC), and reduce out-of-pocket (OOP) payments that threaten financial protection and equity. (Ketevan Glonti, 2013)

Despite important gains - particularly in routine immunization, maternal and child health, and the introduction of strategic purchasing pilots - Uzbekistan continues to face high OOP payments, gaps in financial protection, underdeveloped quality assurance, and pronounced rural–urban disparities. The burden of non-communicable diseases (NCDs) has risen sharply, with ischemic heart disease, stroke, and cancers now the leading causes of mortality. These epidemiological pressures, combined with infrastructure and workforce constraints, have exposed the limitations of traditional, hospital-centric models and underscored the need for innovative, data-driven, and patient-centered approaches. (World Health Organization, 2022)

In this context, digital health—including telemedicine platforms, electronic health records, and e-referral systems—has emerged as a promising instrument for health system improvement. The government's "Concept on Health Development 2019–2025" and subsequent strategies have prioritized digitalization as a lever for expanding access, improving quality, and enhancing efficiency. Early pilots in Syrdarya region, including telemedicine clinics, e-referral systems, and digital queue management, represent a notable inflection point in the country's reform trajectory. (World Health Organization, 2022)

However, the digital health agenda is nascent and faces significant barriers: legacy paper-based systems, fragmented data flows, uneven facility readiness, limited digital literacy, and the absence of robust governance and interoperability standards. Against this backdrop, this thesis offers an integrated theoretical examination of health system transformation in

Uzbekistan, anchored in the specificities of the Uzbek case and informed by comparative regional and international experience.

1.2 Problem Statement and Research Questions

Problem Statement:

Uzbekistan's health system faces persistent challenges in healthcare access, service efficiency, and equity - particularly in rural areas and among vulnerable populations. Despite ongoing reforms and digital health pilots, gaps remain in infrastructure, coordination, and responsiveness. Telemedicine platforms offer a promising solution, but their design must be grounded in the lived experiences and needs of both patients and healthcare providers. This thesis investigates how a telemedicine platform can be developed and tested to address these challenges, based on direct input from users and providers within the Uzbek health system.

Research Questions:

- What are the main barriers to healthcare access as perceived by patients in Uzbekistan, and how can digital tools address these barriers?
- What are the key service delivery inefficiencies experienced by doctors, and what features would improve their workflow and patient management through a telemedicine platform?
- What functional and technical requirements should a telemedicine platform meet to be usable, acceptable, and effective for both patients and providers in Uzbekistan?
- How do users (patients and doctors) respond to a prototype telemedicine platform, and what does their feedback reveal about its potential to improve access and efficiency?

1.3 Objectives of the Thesis

- To identify key barriers to healthcare access and service delivery efficiency in Uzbekistan through empirical surveys of patients and healthcare providers.
- To translate user-reported challenges into functional and technical requirements for a telemedicine platform tailored to Uzbekistan's health system context.

- To design and develop a prototype digital portal for patients and doctors that addresses the identified needs and supports improved access and workflow efficiency.
- To evaluate the usability, acceptability, and perceived effectiveness of the telemedicine platform through pilot testing with real users.
- To contribute to the theoretical and practical understanding of user-centered digital health design in post-Soviet health systems undergoing reform.

2. Healthcare System in Uzbekistan

2.1 Historical Background and Current Organization

Uzbekistan's health system is a product of its Soviet legacy, shaped by the Semashko model—a highly centralized, state-funded, and state-delivered system emphasizing hospital-based care, vertical disease programs, and quantitative targets for inputs such as beds and staff (Bernd Rechel M. A., 2014). During the Soviet era, health care was nominally free at the point of use, with a focus on communicable disease control, maternal and child health, and the expansion of physical infrastructure. However, the system was characterized by inefficiencies, underfunding, and a neglect of primary care and quality assurance (Bernd Rechel E. R., 2014).

Following independence in 1991, Uzbekistan faced the dual challenge of economic transition and health system reform. The early 1990s saw a sharp contraction in public spending, a decline in health indicators, and the emergence of informal payments and private provision (Bernd Rechel M. A., 2014). The government responded with a series of reforms aimed at rationalizing hospital capacity, strengthening PHC, and introducing new financing mechanisms.

Key Reform Milestones:

- **1996 Law on Health Protection:** Established the legal basis for a state-guaranteed benefits package, including primary care, emergency care, and care for "socially significant and hazardous" conditions (Bernd Rechel M. A., 2014).
- **1998 Presidential Decree:** Launched a state program for health system reform, prioritizing PHC, emergency care, and the development of the private sector (Bernd Rechel M. A., 2014).
- **2000s–2010s:** Implementation of World Bank and ADB-supported projects (Health I, II, III; Woman and Child Health Development) focused on PHC infrastructure, maternal and child health, and financing reforms (Bernd Rechel M. A., 2014). (The World Bank, 2025)

- **2018–2021:** Adoption of the "Concept on Health Development 2019–2025" (Presidential Decree 5590), establishment of the SHIF, and piloting of strategic purchasing and digital health initiatives in Syrdarya (The World Bank, 2025).

Current Organization:

The health system is organized into three administrative tiers:

- **National (Republican) Level:** Ministry of Health, national research institutes, tertiary hospitals, and regulatory agencies.
- **Regional (Viloyat) Level:** Regional health authorities, multi-specialty hospitals, and specialized centers.
- **District/City Level:** District medical unions, central hospitals, family polyclinics, and rural physician points (Bernd Rechel M. A., 2014).

The MoH remains the principal steward, with direct management of national-level institutions and regulatory oversight of regional and district facilities. The private sector, while growing, is still limited in scope and subject to strict licensing and regulatory controls (The World Bank, 2025).

Service Delivery Model:

- **Primary Care:** Delivered through family doctor points, polyclinics, and rural physician posts. PHC reforms have aimed to shift from specialist-dominated polyclinics to generalist-led family medicine, but progress has been uneven (Bernd Rechel A. S., 2023).
- **Secondary and Tertiary Care:** Provided by district, regional, and national hospitals, with a legacy of overcapacity and fragmentation. Recent reforms aim to consolidate specialized centers into multi-specialty general hospitals (The World Bank, 2025)
- **Emergency Care:** A well-developed network of emergency departments often better equipped than other public facilities, but subject to overload due to gaps in PHC and benefits coverage (World Health Organization, 2022).

Governance and Regulation:

Regulation is highly centralized, with the MoH and Cabinet of Ministers responsible for policy, financing, and oversight. Local authorities have limited autonomy, and dual accountability structures persist (Bernd Rechel M. A., 2014). The regulatory framework for private providers and pharmaceuticals is evolving, with recent efforts to strengthen licensing, quality assurance, and market surveillance (The World Bank, 2025).

2.2 Key Challenges in Access, Quality, and Financing

Access:

Despite improvements in health indicators, access to quality care remains uneven. Rural-urban disparities are pronounced, with rural areas facing shortages of qualified staff, inadequate infrastructure, and limited access to diagnostics and medicines (Bernd Rechel A. S., 2023). For example, only 57% of PHC facilities reported basic water services and 26% basic sanitation in 2020, reflecting broader readiness constraints (World Health Organization, 2022).

Geographical access is further constrained by the distribution of facilities and workforce. While the number of physicians per 100,000 population has increased to 276 by 2019, and nurse density to ~1,100 per 100,000, these resources are concentrated in urban centers. Outward migration of health professionals, especially to Russia and Kazakhstan, exacerbates shortages in underserved areas. (Bernd Rechel E. R., 2014)

Quality:

Quality of care is undermined by several factors:

- **Legacy of Input-Based Planning:** The Soviet system prioritized quantitative targets (beds, staff) over outcomes, with limited attention to quality assurance, patient safety, or evidence-based practice (Ketevan Glonti, 2013).
- **Outdated Clinical Protocols:** Many facilities rely on outdated or poorly adapted clinical guidelines, with limited adoption of international best practices (Bernd Rechel M. A., 2014).

- **Weak Monitoring and Evaluation:** Data systems are fragmented, paper-based, and focused on structural indicators rather than process or outcome measures. Quality improvement initiatives are limited in scope and sustainability (The World Bank, 2025)

- **Patient Experience:** User experience is often neglected, with long waiting times, limited appointment systems, and a lack of patient-centered care models (Bernd Rechel M. A., 2014).

Financing:

The health financing system is characterized by:

- **Low Public Spending:** Public spending on health was ~2.3% of GDP in 2019, above the Central Asia average but well below the WHO European Region average (~5%) (World Health Organization, 2022).

- **High Out-of-Pocket Payments:** OOP payments accounted for 57.7% of current health expenditure in 2019, exposing households to financial hardship and catastrophic health spending. In 2020, 18% of households reported cost-related nonuse of care (World Health Organization, 2022)

- **Fragmented Pooling and Purchasing:** Budgeting is historically incremental, with limited strategic purchasing or risk pooling. The SHIF pilot in Syrdarya aims to introduce capitation and DRG-based payments, but implementation is nascent and faces capacity constraints (The World Bank, 2025).

- **Limited Benefits Coverage:** The basic benefits package covers primary and emergency care, but excludes much of secondary/tertiary care and outpatient medicines, amplifying OOP and incentivizing emergency use (The World Bank, 2025).

Equity and Financial Protection (Bernd Rechel M. A., 2014)**:**

- **Vulnerable Groups:** While the benefits package includes provisions for vulnerable groups (e.g., children, pregnant women, people with disabilities), eligibility criteria are not directly linked to income, and coverage gaps persist.

- **Informal Payments:** Informal payments remain widespread, particularly in secondary and tertiary care, undermining equity and trust in the system.

2.3 Health Outcomes and Risk Profile

Demographic and Epidemiological Trends (WHO Regional Office for Europe, 2023)**:**

- **Population:** 33.3 million in 2020, with a young age structure but a growing burden of NCDs.

- **Life Expectancy:** Increased from 67.2 years in 2000 to 73.9 years in 2016, with a gender gap of 4.7 years.

- **Maternal and Infant Mortality:** Maternal mortality declined from 41 per 100,000 live births in 2000 to 29 in 2017; infant mortality from 51.8 per 1,000 in 2000 to 15.6 in 2019.

- **NCDs:** Account for 84% of all deaths in 2018, with ischemic heart disease and stroke as leading causes.

- **Risk Factors:** High systolic blood pressure (30.7%), dietary risks (28.2%), and high fasting plasma glucose (20.2%) are dominant contributors to mortality.

Service Coverage and UHC Index (World Health Organization, 2022)**:**

- **UHC Service Coverage Index:** Improved from 56 (2000) to 71 (2019), below OECD/EU comparators but signaling progress.

- **Immunization:** High coverage rates for childhood vaccines, but challenges remain in TB, HIV/AIDS, and MDR-TB.

Social Determinants (The World Bank Group, 2021)**:**

- **Poverty:** Poverty headcount ratio declined from 14.1% in 2013 to 11.4% in 2018, but disparities persist.

- **Water and Sanitation:** Infrastructure needs extensive rehabilitation, with poor access in many areas.

3. Digitalization & Internet Infrastructure

3.1 Introduction: Digital Health as a Systemic Enabler

Digital transformation is now recognized as a foundational enabler of health system reform, not merely a technical add-on. In Uzbekistan, the digital health agenda is explicitly linked to the broader goals of universal health coverage (UHC), equity, efficiency, and quality improvement. The government's "Digital Uzbekistan 2030" strategy and the National Health System Strategy 2030 both position digitalization as a cross-cutting priority, aiming to create robust governance structures, integrated platforms, and the necessary broadband and ICT infrastructure to support health system modernization. (The World Bank, 2025)

However, the digital health landscape in Uzbekistan is shaped by the legacy of Soviet-era information systems, the rapid but uneven expansion of internet and mobile technologies, and persistent gaps in digital literacy and organizational readiness. This section provides a comprehensive analysis of the current state, challenges, and opportunities for digitalization and internet infrastructure in Uzbekistan's health sector.

3.2 Post-2016 Reforms: Infrastructure, Connectivity, and Policy (The World Bank, 2025)

Since 2016, Uzbekistan has embarked on an ambitious digital transformation agenda, with health as a priority sector. The "Digital Uzbekistan 2030" strategy sets out to strengthen digital governance, create integrated information systems, and expand broadband connectivity across the country. The National Health System Strategy 2030 further articulates the role of digital health as a driver of system-wide reform, emphasizing the need for interoperable platforms, data-driven decision-making, and patient empowerment.

By 2023, up to 81% of public medical organizations' legal entities had been provided with internet connectivity. However, the quality and reliability of internet access vary significantly, especially in rural areas. The phased installation of local area networks (LANs) in healthcare facilities is ongoing. Despite these advances, more than 50% of public medical organizations still lack LANs beyond administrative offices, and the estimated need for computers ranges from 30,000 to 65,000 units.

The lack of adequate ICT infrastructure is a critical bottleneck for digital health. Many facilities are underequipped, with insufficient computers, outdated hardware, and no dedicated ICT maintenance staff. The absence of robust data centers further limits the scalability of national digital health systems.

3.3 National and Local Health Information Systems

Uzbekistan's health sector is characterized by a proliferation of more than 35 unconnected nationwide IT e-health systems. These systems operate in silos, with no integration or common master data usage among Ministry of Health (MoH) information systems. There is no national patient registry, no unified registry of healthcare providers, and no standardized nomenclature or classifiers. As a result, integration of IT systems across facilities is not feasible, and data exchange is severely constrained. (WHO Regional Office for Europe, 2023)

At the facility level, only a limited number of organizations use Patient Administration Systems (PAS), Electronic Medical Records (EMR), or Hospital Information Systems (HIS). Under MoH initiatives, two PAS/EMR systems have been piloted in Syrdarya region, and about five local PAS/HIS vendors provide services to public and private clinics and hospitals. However, these systems are not interoperable, and most medical records remain paper-based, scattered across facilities, and inaccessible for longitudinal patient care. (The World Bank, 2025)

3.4 Digital Litracy and Adoption Rates (WHO Regional Office for Europe, 2023)

Digital literacy among health professionals is highly variable. While younger staff and those in urban centers are increasingly proficient with computers and digital tools, many older providers lack basic ICT skills. Continuous IT training is not yet institutionalized, and there is no national strategy for digital capacity building in the health workforce. This gap is compounded by the lack of standardized digital workflows and the persistence of paper-based documentation, which consumes significant provider time and undermines efficiency.

Patient digital literacy is also uneven, with urban populations more likely to use digital health services than rural residents. The absence of a national patient portal or unified engagement platform limits opportunities for patients to access their health records, book appointments, or participate in their care. Where electronic booking systems exist, their use is limited and not widely spread. As a result, patients are not “in the driver’s seat” of their health journey, and digital health remains largely provider-centric.

The success of digital health initiatives depends not only on technical infrastructure but also on organizational readiness and change management. In Uzbekistan, digitalization planning and management are currently centralized within the MoH and its IT-focused company, UZINFOCOM. There is no independent body tasked with overseeing digital health development, and the process is often non-transparent to external stakeholders, including hospitals, clinics, professional associations, and patient groups. This lack of inclusive governance hinders buy-in and slows the pace of digital transformation. (The World Bank, 2025)

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Anhang

Füge hier alle Anlagen hinzu, auf die du in deiner Arbeit verweist.

Du kannst die lesende Person im Fließtext auf Dokumente im Anhang aufmerksam machen, indem du nach einer Aussage in Klammern ‚siehe Anhang' schreibst.

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Anhang laut APA-Richtlinien

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